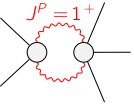
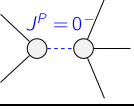
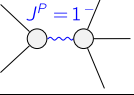
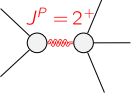
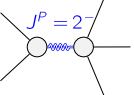


$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$
$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\Upsilon_1 + \Upsilon_2 k^2$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{2} (\Upsilon_3 - \Upsilon_4 k^2)$	$\frac{-\Upsilon_5 k^2 - 2 \binom{(2)}{\kappa_2}}{2 \sqrt{2}}$	0	0	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{-\Upsilon_5 k^2 - 2 \binom{(2)}{\kappa_2}}{2 \sqrt{2}}$	$\frac{1}{2} (\Upsilon_6 k^2 + 2 \binom{(2)}{\kappa_1})$	0	0	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{1^+}^{\#1} \dagger^\alpha$	0	0	$\frac{1}{6} (\Upsilon_7 + 3 \Upsilon_8 k^2)$	$\frac{\Upsilon_7 + 3 \Upsilon_8 k^2}{3 \sqrt{2}}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{\Upsilon_7 + 3 \Upsilon_8 k^2}{3 \sqrt{2}}$	$\frac{1}{3} (\Upsilon_7 + 3 \Upsilon_8 k^2)$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^+}^{\#1}$
$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{2} (\Upsilon_7 + 3 \Upsilon_9 k^2)$	0	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$
$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{2} (\Upsilon_7 + \Upsilon_{10} k^2)$	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^+}^{\#2}$	$\mathcal{K}_{2^+}^{\#1}$

$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$
$\mathcal{J}_{0^+}^{\#1} \dagger$	0	0	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(0^+)}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_6 k^2 + 2 \binom{(2)}{\kappa_1}}{2 \text{Det}(1^+)}$	$\frac{\Upsilon_5 k^2 + 2 \binom{(2)}{\kappa_2}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_5 k^2 + 2 \binom{(2)}{\kappa_2}}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\Upsilon_3 - \Upsilon_4 k^2}{2 \text{Det}(1^+)}$	0	0	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$	0	0	$\frac{1}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$
$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$
$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^+)}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#2}$	$\mathcal{J}_{2^+}^{\#1}$

Abbreviations used in matrices

$\Upsilon_1 == \binom{(2)}{\kappa_1} - \binom{(2)}{\kappa_2} \&\& \Upsilon_2 == \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} \&\& \Upsilon_3 == 2 \binom{(2)}{\kappa_1} - \binom{(2)}{\kappa_2} \&\&$ $\Upsilon_4 == 3 \binom{(4)}{\kappa_1} - 4 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_2} + 2 \binom{(4)}{\kappa_3} \&\& \Upsilon_5 == 2 \binom{(4)}{\kappa_{16}} - \binom{(4)}{\kappa_4} \&\& \Upsilon_6 == 8 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 9 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \&\&$ $\Upsilon_7 == 2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2} \&\& \Upsilon_8 == 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_2} \&\& \Upsilon_9 == \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \&\&$ $\Upsilon_{10} == 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \&\& \text{Det}(0^-) == k^2 (\binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}}) + \binom{(2)}{\kappa_1} - \binom{(2)}{\kappa_2} \&\& \text{Det}(1^+) ==$ $\frac{1}{8} (-k^4 (-64 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 18 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_2} - 54 \binom{(4)}{\kappa_2}^2 + 12 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} - 30 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_3} + 4 \binom{(4)}{\kappa_3}^2 - 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_4} + 6 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_4} +$ $4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + \binom{(4)}{\kappa_4}^2 + 6 \binom{(4)}{\kappa_1} (8 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 9 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) - 8 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_{16}} - 3 (5 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3}) + \binom{(4)}{\kappa_4})) +$ $4 (\binom{(2)}{\kappa_1} - \binom{(2)}{\kappa_2}) (2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2}) + 2 k^2 (2 (-3 \binom{(4)}{\kappa_1} + 12 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 12 \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} +$ $(-8 \binom{(4)}{\kappa_{15}} - 7 \binom{(4)}{\kappa_{16}} + 9 \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_2})) \&\& \text{Det}(2^+) == \frac{1}{2} (3 k^2 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2}) + 2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2})$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha{}_\alpha{}^\beta == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == 0$	
Total # constraint(s):	4		
Unresolved pole(s)	# d.o.f.	Pole structure(s)	
	6	$-k^4 (-64 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 18 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_2} - 54 \binom{(4)}{\kappa_2}^2 + 12 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} - 30 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_3} +$ $4 \binom{(4)}{\kappa_3}^2 - 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_4} + 6 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_4} + 4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + \binom{(4)}{\kappa_4}^2 + 6 \binom{(4)}{\kappa_1} (8 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 9 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) -$ $8 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_{16}} - 3 (5 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3}) + \binom{(4)}{\kappa_4})) + 8 \binom{(2)}{\kappa_1}^2 - 4 \binom{(2)}{\kappa_1} \binom{(2)}{\kappa_2} - 4 \binom{(2)}{\kappa_2}^2 +$ $2 k^2 (2 (-3 \binom{(4)}{\kappa_1} + 12 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 12 \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} + (-8 \binom{(4)}{\kappa_{15}} - 7 \binom{(4)}{\kappa_{16}} + 9 \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_2}))$	
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{-\binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2}}{\binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}}}$	$\frac{1}{-\binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}}$
	3	$-\frac{2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2}}{6 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{16}} - 6 \binom{(4)}{\kappa_2}}$	$-\frac{2}{3 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_2})}$
	5	$-\frac{2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2}}{3 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2})}$	$\frac{2}{3 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2})}$
	5	$-\frac{2 \binom{(2)}{\kappa_1} + \binom{(2)}{\kappa_2}}{2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}}$	$-\frac{2}{2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		